Online Train Reservation System

Aim

To design and develop an Online Train Reservation System that allows passengers to register, log in, search for trains, check seat availability, book tickets, make payments, generate PNR numbers, check PNR status, print tickets, and cancel tickets with refunds. The administrator can manage train details, fares, seat availability, passenger bookings, payment details, and refund rules.

CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Train Reservation System</title>

  <style>
    * {
      box-sizing: border-box;
      font-family: Arial, sans-serif;
    }

    body {
      margin: 0;
      background: #f2f6fb;
      color: #333;
    }

    header {
      background: #123c69;
      color: white;
      text-align: center;
      padding: 22px;
    }

    nav {
```

```
  background: #24527a;
  text-align: center;
  padding: 12px;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 8px;
  display: inline-block;
}

section {
  padding: 25px 5%;
}

h2 {
  color: #123c69;
  text-align: center;
}

.container {
  max-width: 900px;
  margin: auto;
  padding: 22px;
  background: white;
  border-radius: 8px;
  box-shadow: 0 3px 10px #ddd;
}

input, select, button {
  width: 100%;
  padding: 11px;
  margin: 8px 0;
```

```
  border: 1px solid #ccc;
  border-radius: 5px;
}

button {
  background: #123c69;
  color: white;
  border: none;
  cursor: pointer;
}

button:hover {
  background: #2874a6;
}

.train-list {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(230px, 1fr));
  gap: 16px;
}

.train-card {
  background: white;
  padding: 18px;
  border-radius: 8px;
  box-shadow: 0 3px 10px #ddd;
}

.train-card h3 {
  color: #2874a6;
}

.message {
  padding: 12px;
```

```
  background: #e8f5e9;
  margin-top: 12px;
  border-radius: 5px;
  display: none;
}

.table-wrap {
  overflow-x: auto;
}

table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 15px;
}

th, td {
  padding: 10px;
  border: 1px solid #ddd;
  text-align: left;
}

th {
  background: #123c69;
  color: white;
}

.passenger {
  border: 1px solid #ddd;
  border-radius: 5px;
  padding: 12px;
  margin: 10px 0;
}
```

```
    footer {
      background: #123c69;
      color: white;
      text-align: center;
      padding: 20px;
    }

    @media (max-width: 600px) {
      section {
        padding: 18px 3%;
      }
    }

    @media print {
      header, nav, section:not(#ticket), footer,
      #ticket button {
        display: none !important;
      }

      #ticket {
        display: block;
      }
    }
  </style>
</head>

<body>

<header>
  <h1>🚆 Online Train Reservation System</h1>
  <p>Search Trains | Book Tickets | Check PNR Status</p>
</header>

<nav>
```

```
    <a href="#home">Home</a>
    <a href="#login">Login/Register</a>
    <a href="#search">Search Trains</a>
    <a href="#booking">Book Ticket</a>
    <a href="#pnr">PNR Status</a>
    <a href="#history">My Bookings</a>
    <a href="#admin">Admin</a>
</nav>

<section id="home">
    <div class="container">
        <h2>Welcome to Railway Reservation</h2>
        <p>Search trains by source, destination, and journey date.
          Select a class, enter passenger details, and generate a
          demonstration ticket with a PNR number.</p>
    </div>
</section>

<section id="login">
    <div class="container">
        <h2>Passenger Registration</h2>

        <form id="registerForm">
            <input id="regName" placeholder="Full Name" required>
            <input id="regEmail" type="email" placeholder="Email Address" required>
            <input id="regPassword" type="password" minlength="6"
                placeholder="Create Password" required>
            <button type="submit">Register</button>
        </form>

        <h2>Passenger Login</h2>

        <form id="loginForm">
            <input id="loginEmail" type="email" placeholder="Email Address" required>
```

```
        <input id="loginPassword" type="password" placeholder="Password" required>
        <button type="submit">Login</button>
      </form>

      <p id="loginMessage" class="message"></p>
    </div>
</section>

<section id="search">
    <div class="container">
      <h2>Search Trains</h2>

      <form id="searchForm">
        <label>Origin Station</label>
        <input id="source" placeholder="Example: Chennai" required>

        <label>Destination Station</label>
        <input id="destination" placeholder="Example: Bengaluru" required>

        <label>Journey Date</label>
        <input id="journeyDate" type="date" required>

        <button type="submit">Search Available Trains</button>
      </form>
    </div>
</section>

<section id="trains">
    <h2>Available Trains</h2>
    <div class="train-list" id="trainList"></div>
</section>

<section id="booking">
    <div class="container">
```

```
<h2>Book Train Ticket</h2>

<form id="bookingForm">
  <label>Select Train</label>
  <select id="trainSelect" required></select>

  <label>Journey Date</label>
  <input id="bookDate" type="date" required>

  <label>Travel Class</label>
  <select id="travelClass" required>
    <option value="Sleeper">Sleeper</option>
    <option value="3AC">3 AC</option>
    <option value="2AC">2 AC</option>
    <option value="1AC">1 AC</option>
  </select>

  <label>Number of Passengers</label>
  <select id="passengerCount" onchange="createPassengerFields()">
    <option value="1">1 Passenger</option>
    <option value="2">2 Passengers</option>
    <option value="3">3 Passengers</option>
    <option value="4">4 Passengers</option>
    <option value="5">5 Passengers</option>
    <option value="6">6 Passengers</option>
  </select>

  <div id="passengerFields"></div>

  <label>Payment Method</label>
  <select id="paymentMethod" required>
    <option value="Online Demo">Online Payment (Demo)</option>
    <option value="Offline">Offline Payment (Demo)</option>
  </select>
```

```
        <button type="submit">Pay Demo Fare and Book</button>
      </form>

      <p id="bookingMessage" class="message"></p>
    </div>
</section>

<section id="pnr">
  <div class="container">
    <h2>Check PNR Status</h2>

    <form id="pnrForm">
      <input id="pnrInput" placeholder="Enter PNR Number" required>
      <button type="submit">Check Status</button>
    </form>

    <div id="pnrResult"></div>
  </div>
</section>

<section id="history">
  <div class="container">
    <h2>My Booking History</h2>
    <div class="table-wrap">
      <table>
        <thead>
          <tr>
            <th>PNR</th>
            <th>Train</th>
            <th>Date</th>
            <th>Class</th>
            <th>Fare</th>
            <th>Status</th>
```

```
            <th>Action</th>
          </tr>
        </thead>
        <tbody id="historyTable">
          <tr><td colspan="7">Please log in to view bookings.</td></tr>
        </tbody>
      </table>
    </div>
  </div>
</section>

<section id="ticket">
  <div class="container">
    <h2>Generated Ticket</h2>
    <div id="ticketDetails">No ticket generated yet.</div>
    <button onclick="window.print()">Print Ticket</button>
  </div>
</section>

<section id="admin">
  <div class="container">
    <h2>Administrator Login</h2>

    <form id="adminLogin">
      <input id="adminPassword" type="password"
          placeholder="Admin Password" required>
      <button type="submit">Admin Login</button>
    </form>

    <div id="adminPanel" style="display:none;">
      <h2>Manage Train Details</h2>

      <form id="trainForm">
        <input id="trainName" placeholder="Train Name" required>
```

```
          <input id="trainNumber" placeholder="Train Number" required>
          <input id="adminSource" placeholder="Origin Station" required>
          <input id="adminDestination" placeholder="Destination Station" required>
          <input id="trainTime" type="time" required>
          <input id="trainFare" type="number" min="1" placeholder="Base Fare" required>
          <input id="trainSeats" type="number" min="0" max="500"
              placeholder="Available Seats" required>
          <button type="submit">Add Train</button>
        </form>

        <h2>Existing Trains</h2>
        <div id="adminTrainList"></div>

        <h2>Manage Reservations</h2>
        <div id="adminBookingList"></div>

        <h2>Refund Rules</h2>
        <label>Refund Percentage</label>
        <input id="refundPercent" type="number" min="0" max="100" value="80">
        <button onclick="saveRefundRule()">Save Refund Rule</button>

        <h2>Payment and Booking Report</h2>
        <button onclick="generateReport()">Generate Report</button>
        <div id="report"></div>
      </div>
    </div>
</section>

<footer>
    <p>&copy; 2026 Online Train Reservation System</p>
    <p>Internet Programming Lab Experiment</p>
</footer>

<script>
```

```
let users = [];
let currentUser = null;
let bookings = [];
let nextPNR = 100000001;

let trains = [
  {
    id: 1,
    number: "12601",
    name: "Chennai Express",
    source: "Chennai",
    destination: "Bengaluru",
    time: "06:00",
    fare: 500,
    seats: 40
  },
  {
    id: 2,
    number: "12603",
    name: "Southern Superfast",
    source: "Chennai",
    destination: "Hyderabad",
    time: "08:30",
    fare: 700,
    seats: 30
  },
  {
    id: 3,
    number: "12605",
    name: "City Intercity",
    source: "Bengaluru",
    destination: "Chennai",
    time: "15:00",
    fare: 450,
```

```
    seats: 35
  }
];

let refundPercentage = 80;

function escapeHTML(value) {
  return String(value).replace(/[&<>"']/g, character => ({
    "&": "&amp;",
    "<": "&lt;",
    ">": "&gt;",
    '"': "&quot;",
    "'": "&#39;"
  })[character]);
}

function message(id, text) {
  const element = document.getElementById(id);
  element.textContent = text;
  element.style.display = "block";
}

function getFare(train, travelClass) {
  const multiplier = {
    "Sleeper": 1,
    "3AC": 1.8,
    "2AC": 2.5,
    "1AC": 3.5
  };

  return Math.round(train.fare * multiplier[travelClass]);
}

document.getElementById("registerForm").addEventListener("submit", function(e) {
```

```
  e.preventDefault();

  const name = document.getElementById("regName").value.trim();
  const email = document.getElementById("regEmail").value.trim().toLowerCase();
  const password = document.getElementById("regPassword").value;

  if (users.some(user => user.email === email)) {
    message("loginMessage", "Email already registered.");
    return;
  }

  users.push({name, email, password});
  message("loginMessage", "Registration successful. Please log in.");
  this.reset();
});

document.getElementById("loginForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const email = document.getElementById("loginEmail").value.trim().toLowerCase();
  const password = document.getElementById("loginPassword").value;

  currentUser = users.find(u => u.email === email && u.password === password);

  if (!currentUser) {
    message("loginMessage", "Invalid login details. Please register first.");
    return;
  }

  message("loginMessage", "Welcome, " + currentUser.name + "!");
  renderHistory();
});

function renderTrains(list = trains) {
```

```
  document.getElementById("trainList").innerHTML = list.map(train => `
    <div class="train-card">
      <h3>${escapeHTML(train.name)}</h3>
      <p>Train Number: ${escapeHTML(train.number)}</p>
      <p>Route: ${escapeHTML(train.source)} to ${escapeHTML(train.destination)}</p>
      <p>Departure: ${escapeHTML(train.time)}</p>
      <p>Base Fare: ₹${train.fare}</p>
      <p>Available Seats: ${train.seats}</p>
      <button onclick="selectTrain(${train.id})">Book This Train</button>
    </div>
  `).join("");

  const select = document.getElementById("trainSelect");
  const previous = select.value;

  select.innerHTML = '<option value="">Select a train</option>' +
    trains.map(t =>
      `<option value="${t.id}">${escapeHTML(t.name)} (${escapeHTML(t.number)})</option>`
    ).join("");

  if (trains.some(t => String(t.id) === previous)) {
    select.value = previous;
  }

  renderAdminTrains();
}

function selectTrain(id) {
  document.getElementById("trainSelect").value = String(id);
  document.getElementById("booking").scrollIntoView({behavior: "smooth"});
}

document.getElementById("searchForm").addEventListener("submit", function(e) {
  e.preventDefault();
```

```
  const source = document.getElementById("source").value.trim().toLowerCase();
  const destination = document.getElementById("destination").value.trim().toLowerCase();
  const date = document.getElementById("journeyDate").value;

  if (date < new Date().toISOString().slice(0, 10)) {
    alert("Select today or a future journey date.");
    return;
  }

  document.getElementById("bookDate").value = date;

  const result = trains.filter(t =>
    t.source.toLowerCase() === source &&
    t.destination.toLowerCase() === destination
  );

  document.getElementById("trainList").innerHTML = result.length
    ? result.map(train => `
      <div class="train-card">
        <h3>${escapeHTML(train.name)}</h3>
        <p>Train Number: ${escapeHTML(train.number)}</p>
        <p>${escapeHTML(train.source)} to ${escapeHTML(train.destination)}</p>
        <p>Departure: ${escapeHTML(train.time)}</p>
        <p>Journey Date: ${escapeHTML(date)}</p>
        <p>Base Fare: ₹${train.fare}</p>
        <p>Available Seats: ${train.seats}</p>
        <button onclick="selectTrain(${train.id})">Book This Train</button>
      </div>
    `).join("")
    : "<p>No trains found for this route.</p>";
});

function createPassengerFields() {
```

```
  const count = Number(document.getElementById("passengerCount").value);
  const area = document.getElementById("passengerFields");

  area.innerHTML = Array.from({length: count}, (_, i) => `
    <div class="passenger">
      <h3>Passenger ${i + 1}</h3>
      <input class="p-name" placeholder="Passenger Name" required>
      <input class="p-age" type="number" min="1" max="120"
          placeholder="Age" required>
      <select class="p-gender" required>
        <option value="">Select Gender</option>
        <option>Female</option>
        <option>Male</option>
        <option>Other</option>
      </select>
    </div>
  `).join("");
}

document.getElementById("bookingForm").addEventListener("submit", function(e) {
  e.preventDefault();

  if (!currentUser) {
    alert("Please log in before booking a ticket.");
    return;
  }

  const trainId = Number(document.getElementById("trainSelect").value);
  const train = trains.find(t => t.id === trainId);
  const date = document.getElementById("bookDate").value;
  const travelClass = document.getElementById("travelClass").value;
  const count = Number(document.getElementById("passengerCount").value);

  if (!train || !date || date < new Date().toISOString().slice(0, 10)) {
```

```
    alert("Select a valid train and journey date.");
    return;
}

if (train.seats < count) {
    alert("Not enough seats available.");
    return;
}

const names = [...document.querySelectorAll(".p-name")].map(e => e.value.trim());
const ages = [...document.querySelectorAll(".p-age")].map(e => Number(e.value));
const genders = [...document.querySelectorAll(".p-gender")].map(e => e.value);

if (names.some(n => !n) ||
    ages.some(a => !Number.isInteger(a) || a < 1 || a > 120) ||
    genders.some(g => !g)) {
    alert("Please complete all passenger details.");
    return;
}

const fare = getFare(train, travelClass) * count;
const payment = document.getElementById("paymentMethod").value;

// Demonstration only: no actual payment is processed.
if (!confirm("Confirm demo payment of ₹" + fare + " using " + payment + "?")) {
    return;
}

train.seats -= count;

const booking = {
    pnr: String(nextPNR++),
    email: currentUser.email,
    trainId: train.id,
```

```
      trainName: train.name,
      trainNumber: train.number,
      source: train.source,
      destination: train.destination,
      date,
      travelClass,
      passengers: names.map((name, i) => ({
        name,
        age: ages[i],
        gender: genders[i]
      })),
      fare,
      payment,
      status: "Confirmed"
    };

    bookings.push(booking);

    message("bookingMessage",
      "Demo ticket booked successfully. Your PNR is " + booking.pnr +
      ". Total fare: ₹" + fare);

    showTicket(booking);
    renderHistory();
    renderAdminBookings();
    renderTrains();
    this.reset();
    createPassengerFields();
});

function showTicket(booking) {
    document.getElementById("ticketDetails").innerHTML = `
      <h3>Railway E-Ticket (Demo)</h3>
      <p><strong>PNR:</strong> ${escapeHTML(booking.pnr)}</p>
```

```
      <p><strong>Train:</strong> ${escapeHTML(booking.trainName)}
        (${escapeHTML(booking.trainNumber)})</p>
      <p><strong>Route:</strong> ${escapeHTML(booking.source)} to
        ${escapeHTML(booking.destination)}</p>
      <p><strong>Journey Date:</strong> ${escapeHTML(booking.date)}</p>
      <p><strong>Class:</strong> ${escapeHTML(booking.travelClass)}</p>
      <p><strong>Passengers:</strong>
        ${booking.passengers.map(p => escapeHTML(p.name) +
        " (Age " + p.age + ")").join(", ")}</p>
      <p><strong>Fare:</strong> ₹${booking.fare}</p>
      <p><strong>Status:</strong> ${escapeHTML(booking.status)}</p>
    `;
}

function renderHistory() {
  const table = document.getElementById("historyTable");

  if (!currentUser) {
    table.innerHTML = '<tr><td colspan="7">Please log in to view bookings.</td></tr>';
    return;
  }

  const mine = bookings.filter(b => b.email === currentUser.email);

  table.innerHTML = mine.length ? mine.map(b => `
    <tr>
      <td>${escapeHTML(b.pnr)}</td>
      <td>${escapeHTML(b.trainName)}</td>
      <td>${escapeHTML(b.date)}</td>
      <td>${escapeHTML(b.travelClass)}</td>
      <td>₹${b.fare}</td>
      <td>${escapeHTML(b.status)}</td>
      <td>
        ${b.status === "Confirmed"
```

```
            ? `<button onclick="cancelTicket('${b.pnr}')">Cancel</button>
              <button onclick="printTicket('${b.pnr}')">Print</button>`
            : "-"}
        </td>
      </tr>
    `).join("") : '<tr><td colspan="7">No bookings found.</td></tr>';
}

document.getElementById("pnrForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const pnr = document.getElementById("pnrInput").value.trim();
  const booking = bookings.find(b => b.pnr === pnr);

  document.getElementById("pnrResult").innerHTML = booking
    ? `<div class="container">
        <h3>PNR Status</h3>
        <p>PNR: ${escapeHTML(booking.pnr)}</p>
        <p>Train: ${escapeHTML(booking.trainName)}</p>
        <p>Journey Date: ${escapeHTML(booking.date)}</p>
        <p>Status: ${escapeHTML(booking.status)}</p>
       </div>`
    : "<p>PNR not found in this demo.</p>";
});

function cancelTicket(pnr) {
  const booking = bookings.find(b =>
    b.pnr === pnr && b.email === currentUser.email
  );

  if (!booking || booking.status !== "Confirmed") return;

  if (!confirm("Cancel ticket with PNR " + pnr + "?")) return;
```

```
    booking.status = "Cancelled";

    const train = trains.find(t => t.id === booking.trainId);
    if (train) train.seats += booking.passengers.length;

    const refund = Math.round(booking.fare * refundPercentage / 100);

    alert("Demo cancellation completed. Illustrative refund: ₹" + refund +
        ". Actual refunds require payment processing.");

    renderHistory();
    renderAdminBookings();
    renderTrains();
}

function printTicket(pnr) {
    const booking = bookings.find(b =>
        b.pnr === pnr && b.email === currentUser.email
    );

    if (!booking) return;

    showTicket(booking);
    window.print();
}

document.getElementById("adminLogin").addEventListener("submit", function(e) {
    e.preventDefault();

    // Demo password only; real authentication must be server-side.
    if (document.getElementById("adminPassword").value === "admin123") {
        document.getElementById("adminPanel").style.display = "block";
        renderAdminTrains();
        renderAdminBookings();
```

```
    alert("Demo admin login successful.");
  } else {
    alert("Incorrect admin password.");
  }
});

document.getElementById("trainForm").addEventListener("submit", function(e) {
  e.preventDefault();

  trains.push({
    id: Math.max(0, ...trains.map(t => t.id)) + 1,
    name: document.getElementById("trainName").value.trim(),
    number: document.getElementById("trainNumber").value.trim(),
    source: document.getElementById("adminSource").value.trim(),
    destination: document.getElementById("adminDestination").value.trim(),
    time: document.getElementById("trainTime").value,
    fare: Number(document.getElementById("trainFare").value),
    seats: Number(document.getElementById("trainSeats").value)
  });

  renderTrains();
  this.reset();
  alert("Train added to the demo system.");
});

function renderAdminTrains() {
  document.getElementById("adminTrainList").innerHTML = trains.map(train => `
    <div class="train-card">
      <h3>${escapeHTML(train.name)}</h3>
      <p>Number: ${escapeHTML(train.number)}</p>
      <p>${escapeHTML(train.source)} to ${escapeHTML(train.destination)}</p>
      <p>Fare: ₹${train.fare} | Seats: ${train.seats}</p>
      <button onclick="editTrain(${train.id})">Edit Fare/Seats</button>
      <button onclick="removeTrain(${train.id})">Remove Train</button>
```

```
      </div>
    `).join("");
}

function editTrain(id) {
    const train = trains.find(t => t.id === id);
    if (!train) return;

    const fareInput = prompt("Enter new base fare:", train.fare);
    const seatsInput = prompt("Enter new available seat count:", train.seats);

    if (fareInput === null || seatsInput === null) return;

    const fare = Number(fareInput);
    const seats = Number(seatsInput);

    if (!Number.isFinite(fare) || fare <= 0 ||
        !Number.isInteger(seats) || seats < 0) {
        alert("Enter a valid fare and seat count.");
        return;
    }

    train.fare = fare;
    train.seats = seats;

    renderTrains();
    renderAdminTrains();
}

function removeTrain(id) {
    if (bookings.some(b => b.trainId === id && b.status === "Confirmed")) {
        alert("Cancel active bookings before removing this train.");
        return;
    }
```

```
    if (!confirm("Remove this train?")) return;

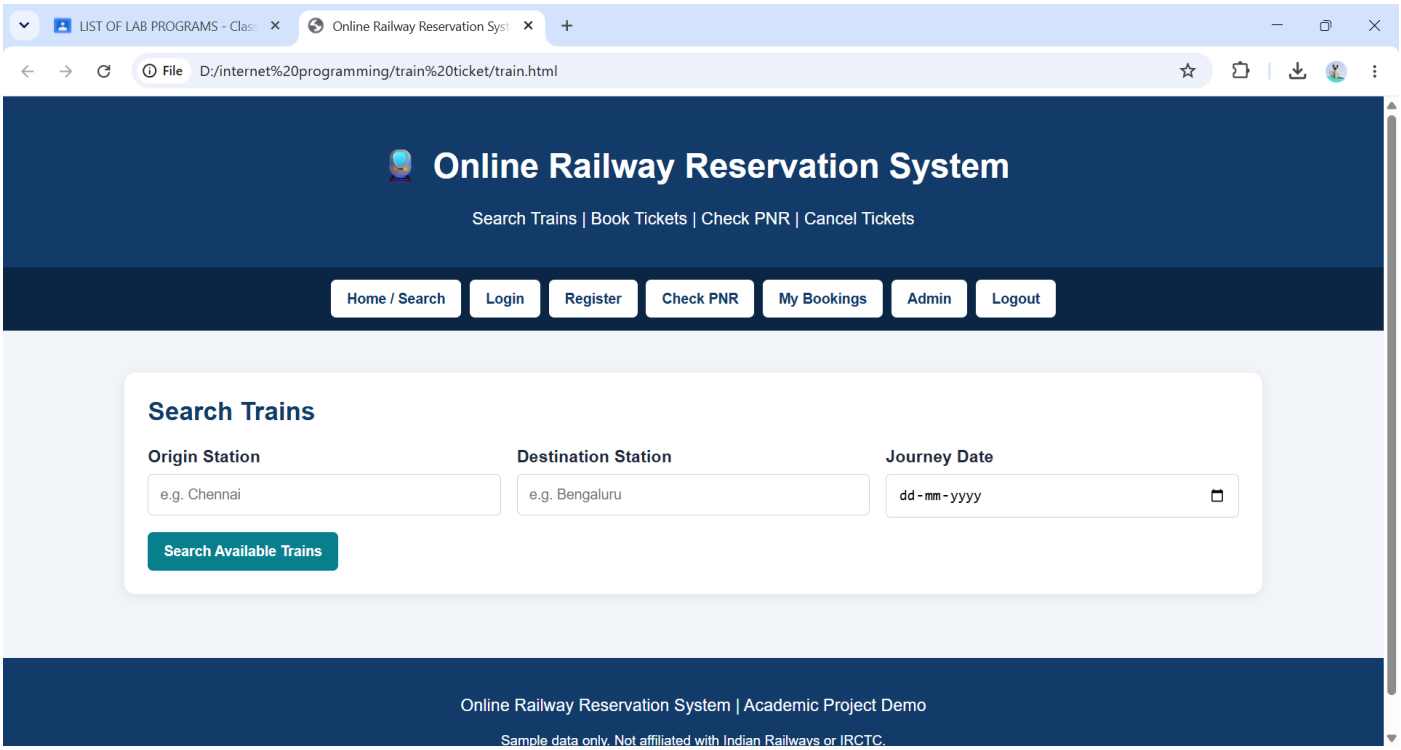
    trains = trains.filter(t => t.id !== id);
    renderTrains();
  }

  function renderAdminBookings() {
    const area = document.getElementById("adminBookingList");

    area.innerHTML = bookings.length ? bookings.map(b => `
      <div class="train-card">
        <p>PNR: ${escapeHTML(b.pnr)}</p>
        <p>Passenger account: ${escapeHTML(b.email)}</p>
        <p>Train: ${escapeHTML(b.trainName)}</p>
        <p>Date: ${escapeHTML(b.date)}</p>
        <p>Fare: ₹${b.fare}</p>
        <p>Payment method: ${escapeHTML(b.payment)}</p>
        <p>Status: ${escapeHTML(b.status)}</p>
        ${b.status === "Confirmed"
          ? `<button onclick="adminCancelTicket('${b.pnr}')">Cancel Ticket</button>`
          : ""}
      </div>
    `).join("") : "<p>No bookings yet.</p>";
  }

  function adminCa
```

RESULT


LIST OF LAB PROGRAMS - Class
Online Railway Reservation Syst
File
D:/internet%20programming/train%20ticket/train.html
Online Railway Reservation System
Search Trains | Book Tickets | Check PNR | Cancel Tickets
Home / Search
Login
Register
Check PNR
My Bookings
Admin
Logout
Search Trains
Origin Station
e.g. Chennai
Destination Station
e.g. Bengaluru
Journey Date
dd-mm-yyyy
Search Available Trains
Online Railway Reservation System | Academic Project Demo
Sample data only. Not affiliated with Indian Railways or IRCTC.